



**CS 353 - Database Systems
Spring 2026**

Project Proposal

Group 2

Uğurcan Doğruçam 22302777

Eren Sucuoğlu 22302576

Talha Safa Küçük 22302802

Ahmet Güneyli 22302556

Hidayet Başer 22302003

Date: 19.02.2026

Table of Contents

1. Problem Description.....	3
1.1. Overview.....	3
1.2. Purpose.....	3
1.3. Database Necessity.....	3
1.4. Target Users.....	3
2. System Requirements.....	3
2.1. Functional Requirements.....	3
2.1.1. Multi-Criteria Recipe Filtering.....	3
2.1.2. Dynamic Ingredient Scaling.....	4
2.1.3. Real-Time Inventory Integration.....	4
2.1.4. Taxonomy-Based Substitution.....	4
2.1.5. Automated Cart Generation.....	4
2.1.6. Standardized Recipe Creation.....	4
2.1.7. Royalty Analytics Dashboard.....	4
2.1.8. Kitchen Challenge Management.....	4
2.1.9. Social Feedback & Logging.....	4
2.1.10. Custom Meal Lists.....	4
2.1.11. Administrative Taxonomy Mapping.....	5
2.1.12. Reporting and Insights.....	5
2.2. Non-Functional Requirements.....	5
2.2.1. Data Integrity.....	5
2.2.2. Security.....	5
2.2.3. Performance.....	5
2.2.4. Scalability.....	5
3. System Limitations.....	6
3.1. No Real Payment Processing.....	6
3.2. No Real-Time Delivery Tracking.....	6
3.3. Static Freshness Data.....	6
3.4. No Video Streaming.....	6
3.5. Single Currency.....	6
3.6. No Automatic Nutritional Analysis.....	6
4. E/R Diagram.....	7

1. Problem Description

The **Culinary Discovery & Meal Kit Marketplace** is a web-based application designed to bridge the gap between home cooks, professional chefs, and local ingredient suppliers.

1.1 Overview: In the current culinary landscape, home cooks often struggle to find reliable recipes that match their dietary needs and then face the secondary challenge of sourcing specific, high-quality ingredients. Simultaneously, local suppliers lack a direct digital channel to consumers, and chefs struggle to monetize their digital content effectively.

1.2 Purpose: This system solves these problems by creating a unified ecosystem. It allows users to discover recipes, automatically generates shopping carts based on serving sizes ("Shop This Meal"), and routes orders to local suppliers.

1.3 Database Necessity: A relational database is essential for managing the complex relationships among dynamic inventory (ingredients sourced from various suppliers with fluctuating prices), recipe structures (hierarchical ingredient structures and scaling logic), user interactions (ratings, challenges), and financial tracking (past purchases).

1.4 Target Users:

- **Home Cooks:** Browse recipes, purchase meal kits, and participate in cooking challenges.
- **Verified Chefs:** Create content and earn royalties.
- **Local Suppliers:** Manage inventory and fulfill orders.
- **Admins:** Oversee the platform, verify users, and manage content.

2. System Requirements

2.1. Functional Requirements (FR):

These define the specific behaviors and features the system must provide:

2.1.1. Multi-Criteria Recipe Filtering:

The system will allow Home Cooks to filter recipes based on dietary restrictions (e.g., Keto, Vegan), cooking time, specific ingredients, and minimum chef ratings.

2.1.2. Dynamic Ingredient Scaling:

The system must automatically recalculate all ingredient quantities when a user modifies the serving size (e.g., scaling a recipe from 2 to 6 people).

2.1.3. Real-Time Inventory Integration:

Upon clicking "Shop This Meal," the system will perform a real-time check of Local Suppliers' inventory to ensure all required ingredients are in stock.

2.1.4. Taxonomy-Based Substitution:

If a specific ingredient is unavailable, the system will suggest a substitute based on a predefined ingredient taxonomy (e.g., suggesting "Vine Tomato" if "Roma Tomato" is out of stock).

2.1.5. Automated Cart Generation:

The system will generate a shopping cart containing the exact portions of ingredients required for a selected recipe and serving size.

2.1.6. Standardized Recipe Creation:

The system will provide a standardized interface for Chefs to input recipe metadata, preparation steps, and ingredients selected from a master taxonomy.

2.1.7. Royalty Analytics Dashboard:

The system will provide Verified Chefs with a dashboard to track "cook" actions, user logs, and calculated earned royalties.

2.1.8. Kitchen Challenge Management:

The system will allow users to browse and join time-bound challenges (e.g., "Zero Waste Week") and track their progress based on the qualifying recipes they cook.

2.1.9. Social Feedback & Logging:

The system will allow Home Cooks to rate, review, and log cooked recipes, which will serve as a trigger for royalty updates.

2.1.10. Custom Meal Lists:

Users will be able to create, name, and manage personal collections

of recipes (e.g., "Best for Family Gatherings").

2.1.11. Administrative Taxonomy Mapping:

Admins will be able to map specific ingredients to generic categories (e.g., "Roma Tomato" to the generic "Tomato" entity) for improved search and substitution.

2.1.12. Reporting and Insights:

The system will generate administrative reports on "Trending Ingredients" and "Most Profitable Recipes" based on purchase and log data

2.2. Non-functional Requirements

2.2.1. Data Integrity:

The system must ensure that inventory counts never drop below zero during concurrent purchase attempts.

2.2.2. Security:

User passwords must be stored using strong hashing algorithms (e.g., SHA-256 or bcrypt), and sensitive address data must be protected in the database.

2.2.3. Performance:

Recipe search and filtering queries must return results in under 3 seconds, even with a large recipe dataset.

2.2.4. Scalability:

The database design must support thousands of recipes and transaction records without significant degradation in query performance.

3. System Limitations

Constraints outlining what the system will not be able to do.

3.1. No Real Payment Processing:

The system uses mock balances and does not integrate with real-world banking APIs or credit card processors.

3.2. No Real-Time Delivery Tracking:

The system handles order creation but does not track the physical logistics or GPS locations of delivery drivers.

3.3. Static Freshness Data:

Ingredient expiration and freshness are manually entered by suppliers; the system does not use IoT sensors to detect spoilage.

3.4. No Video Streaming:

The platform will not host live-streaming cooking classes; it is limited to static uploads such as recipe metadata, preparation steps, and ingredients.

3.5. Single Currency:

The platform uses a single currency and does not support multi-currency exchange rates for international suppliers.

3.6. No Automatic Nutritional Analysis:

The system does not chemically analyze recipes for calorie/macro counts; it relies on data entered manually by the Chef.

4. ER Diagram

